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Dimerisation of ethanol to butanol over solid-base catalysts

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Abstract

Ethanol is converted into 1-butanol over alkali earth metal oxides and modified MgO catalysts (1–20% yield). The MgO catalyst exhibits the highest reaction activity and 1-butanol selectivity amongst the catalysts studied. Reaction of various intermediates (acetaldehde, crotonaldehyde, crotylalcohol, butanal) and butanol over MgO (1 bar, 450 °C) revealed that the dimerisation reaction does not proceed primarily through the aldol condensation reaction. The reaction is proposed to proceed through a mechanism (Yang and Meng) in which a C–H bond in the β-position in ethanol is activated by the basic metal oxide, and condenses with another molecule of ethanol by dehydration to form 1-butanol.

Keywords: Magnesium oxide; Base catalysis; Butanol; Ethanol

1. Introduction

Chemical conversions over solid-acid catalysts have received extensive coverage over the decades [1–4]. An example is the conversion of ethanol into higher value chemicals, a reaction that has received much attention in the literature [5,6]. In comparison to solid-acid catalysts, less attention has been devoted to solid-base catalysts, even though high activities and selectivities are often attained for chemical conversions carried out over *liquid* bases for many kinds of reactions.

Many homogeneous catalysed base reactions are known and include reactions such as isomerisation, addition, alkylation and cyclisation [7–10]. Numerous reactions are also known that require a stoichiometric amount of liquid bases [11]. The replacement of liquid bases by *solid-base catalysts* would provide environmental advantages to industrial processes. These include issues relating to decreased reactor corrosion, easier product/catalyst separation and recovery of used catalysts [8].

The use of solid-base catalysts for the synthesis of fine chemicals from alcohols has not been extensively studied [8]. Condensation of various primary alcohols to form higher alcohols is however, known [12–17]. An important industrial process that is used to increase the carbon number of alcohols is the Guerbet reaction [18]. In this reaction, a primary or secondary alcohol reacts with itself or another alcohol to produce a higher alcohol. This reaction has been developed and various types of heterogeneous catalysts for this reaction have been reported in patents [18,19]. Ueda et al. [20,21] have also reported a catalytic reaction for producing higher alcohols using methanol as a building block. In this reaction, carried out over a solid-base metal oxide catalyst, methanol is condensed with other primary or secondary alcohols having a methyl or methylene group at the β-position.

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Surprisingly, the reaction of ethanol to generate 1-butanol, over solid-base catalysts has been little reported in the literature. 1-Butanol is a crucial building block for acrylic acid and acrylic esters and is widely used as a solvent [13–15,22]. It has also found application as an additive to gasoline [22].

Herein, we report the synthesis of 1-butanol from ethanol over a base catalyst. A thorough investigation of the reaction using a variety of possible intermediates suggests that the reaction involves a bimolecular condensation reaction [23].

2. Experimental

2.1. Catalyst preparation

Magnesium oxide catalysts were prepared from MgO 'heavy' (A.R.) (Hopkin and Williams Ltd.). The magnesium oxide was suspended in warm water with stirring and then solidified again by the evaporation of the water. It was then dried at 393 K for 16 h and finally calcined in air at 773 K for 6 h. This catalyst was found to have a BET surface area of 38 m^2/g. CaO and BaO were prepared from the thermal decomposition of calcium hydroxide and barium hydroxide, respectively, at elevated temperature (773 K) for 6 h.

Silica (Aerosil 200) and MgO "lite" (treated in way similar to that of 'heavy' MgO) with BET surface areas of 67 and 54 m^2/g, respectively, were used as supports for alkali, alkaline earth and transition metal oxides. γ-Alumina (BET surface area: 157 m^2/g and pore volume: 0.23 cm^3/g) was also used as a support.

The metal oxide supported catalysts were prepared by an impregnation procedure using an appropriate amount of aqueous metal hydroxide/nitrate solutions (incipient wetness method). The catalysts were dried at 393 K (16 h) and calcined in air at 773 K.

2.2. Apparatus and procedure

The catalytic experiments were carried out in a vertical fixed-bed glass reactor (15 mm i.d.). All reactions were carried out using 0.5 g of catalyst. The catalysts were used in the form of granules (350–850 μm) prepared by pelletising the catalyst powder and crushing to the desired size. The reactor was placed inside a temperature-controlled heating jacket with a thermocouple placed at the centre of the catalyst bed.

The catalyst was activated in flowing oxygen at 773 K for 1 h prior to flushing with nitrogen gas (10 ml/min). Ethanol was fed from a 10 ml syringe using a syringe pump (Sage instruments, model 341B) through a heated carrier gas (nitrogen) line prior to the reactor. Reactions were conducted under nitrogen flow (10 ml/min). The products were collected in a water-cooled receiver and analysed in a gas chromatograph (ZB wax, 60 m × 0.32 mm capillary column; FI Detector). Product identification of the major components was achieved by GC–MS (Finnigan TSQ-7000) and ^{13}C-NMR spectroscopy (Brucker AC 400) and the use of standards. In early experiments, online GC analysis was also performed and CO/CO_2 as well as small amounts of C_1–C_4 gases were detected. These included ethene, ethane, butane and butenes. Minor amounts of C_6 and C_8 compounds, formed by further dimerisation reactions were also formed in the liquid phase; these products can also be prepared from $C_4 + C_2$ and $C_4 + C_4$ reactions of alcohols and the results of these studies will be reported elsewhere.

The ethanol conversion and product yield were calculated as follows:

$$\%\,\text{conversion} = \left(\frac{\text{Moles of ethanol reacted}}{\text{Moles of ethanol introduced}}\right) \times 100$$

and

$$\%\,\text{yield} = \left(\frac{\text{Moles of product}}{\text{Moles of ethanol introduced}}\right) \times 100.$$

Calculations of product yields were based on ethanol reacted and take account of all products produced.

The temperature-programmed desorption (TPD) apparatus consists of a U-shaped quartz reactor housed in a temperature-controlled heating jacket. The signal is measured by a low-temperature dual filament TCD and recorded by a PC equipped with an I/O card.

3. Results and discussion

3.1. Catalyst screening

Ethanol was passed over a range of catalysts at 450 °C (Table 1), and a substantial amount of liquid products were observed. In addition, some gaseous

Table 1
Activities of basic oxides and alkali-loaded Al_2O_3 in the solid-base condensation of ethanol to butanol[a]

Catalyst	Conversion (mol%)	Yield (mol%)					
		Acetal	Butanal	Crotonal	2-Butanol	Butanol	Crotanol
MgO	56.14	0.65	0.78	0.50	0.14	18.39	0.31
CaO	15.16	2.83	2.15	0.32	0.07	1.16	0.06
BaO	21.19	6.56	11.37	–	–	–	–
γ-Al_2O_3	82.29	5.96	19.42	–	–	–	–
10% Na/Al_2O_3	35.52	0.32	2.56	0.06	0.18	0.92	0.05
10% K/Al_2O_3	54.02	1.09	0.75	0.20	0.26	3.42	0.04
10% Cs/Al_2O_3	24.57	0.31	0.31	–	–	0.05	–
10% Mg/SiO_2	18.64	5.04	0.17	0.17	–	0.38	0.05

[a] Acetal: acetaldehyde; butanal: butyraldehyde; crotonal: crotonaldehyde; crotanol: crotylalcohol.

light products were formed by thermal decomposition of a part of the ethanol. The selectivities and activities of the different catalysts were evaluated by GC studies. Product compositions were determined by a combination of GC, GC–MS and ^{13}C-NMR spectroscopy. By this means all of the major products were identified.

MgO showed the highest yield of butanol among the catalysts studied (Table 1). CaO showed the least activity and produced very little butanol. Barium oxide did not show any butanol formation. Although high *activity* was observed over γ-alumina, no butanol was formed over this catalyst. Butanal and acetaldehyde, together with substantial amounts of gaseous products, were the major components observed. This is in accord with the acidic nature of this catalyst. The impregnation of alkali metals on γ-alumina, resulted in the formation of a small amount of butanol. Coverage of the Al_2O_3 by the metal ions resulted in an increase in the amount of butanol produced relative to Al_2O_3 suggesting that the basic sites produced by the metal ions were required for butanol formation. The ethanol conversion of the Al_2O_3 catalysts also showed a general decrease with metal ion coverage.

Attempts were then made to enhance the catalyst selectivity/activity by enhancing the MgO surface area and by addition of metal ions. Dispersion of MgO on a series of metal oxides did not improve the yield of butanol (see an example in Table 1, Mg/SiO_2). Similarly, addition of the basic metal ions Ca and Ba to MgO also did not enhance the catalyst selectivity (Table 2). Thus, even though the basic nature of the MgO was modified by loading calcium and barium on MgO no advantages in terms of catalyst productivity were achieved—little effect on the yield of C_4 products was noted and the yield of butanol decreased significantly. Ueda et al. [20,21] have also reported similar findings for the reaction of ethanol and methanol to form 1-propanol, where they found that while the addition

Table 2
Activities of alkaline earth and transition metals loaded on MgO catalysts in the solid-base condensation of ethanol to butanol[a]

Catalyst	Conversion (mol%)	Yield (mol%)					
		Acetal	Butanal	Crotonal	2-Butanol	Butanol	Crotanol
10% Ba/MgO	34.19	15.35	0.81	0.15	0.41	7.33	0.47
10% Ca/MgO	14.55	18.15	0.83	0.21	3.26	7.58	0.84
10% Zn/MgO	95.61	1.11	6.93	1.79	2.91	2.95	0.03
10% Ce/MgO	70.84	0.56	4.27	0.09	0.00	0.50	0.28
10% Zr/MgO	52.49	6.48	5.65	0.14	0.06	0.29	0.14
10% Pb/MgO	34.18	6.46	3.61	7.51	2.17	1.97	0.81
10% Sn/MgO	57.21	5.62	1.56	0.85	3.95	1.27	0.05
10% Cu/MgO	28.75	18.10	0.70	0.12	0.60	1.80	0.82

[a] Acetal: acetaldehyde; butanal: butyraldehyde; crotonal: crotonaldehyde; crotanol: crotylalcohol.

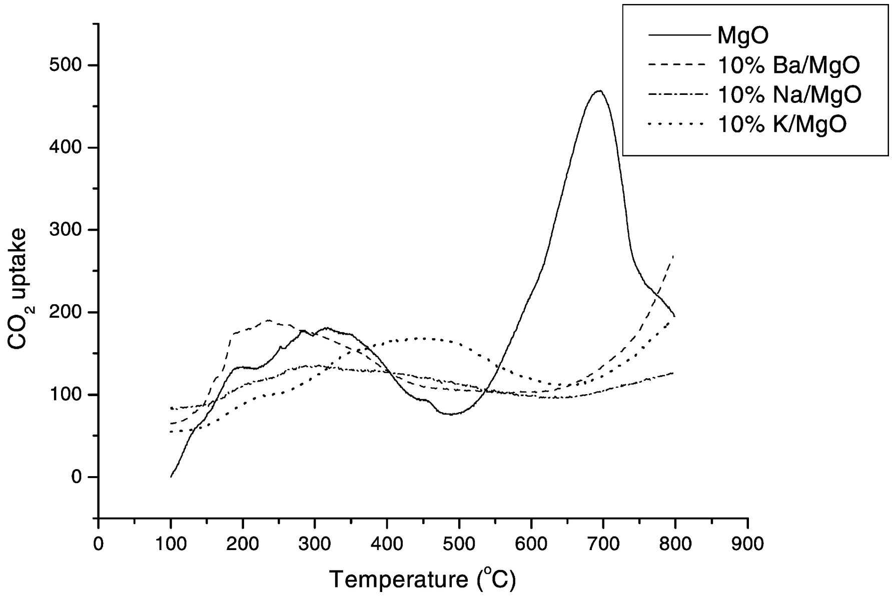


Fig. 1. TPD profiles of some of the catalysts.

of alkali metals increased the basicity of oxygen anions in MgO, condensation rates were decreased.

The addition of a range of transition metals (Zn, Ce, Zr, Pb, Sn) to MgO was also investigated (Table 2). These ions were expected to increase the basic nature of the MgO but here again the metal ion addition did not improve the yield of butanol or other C_4 compounds. Ueda et al. [24] have reported that the surface basicity of MgO was increased by the addition of metal ions when the ionic radius was greater than that of Mg^{2+}.

TPD experiments were performed with CO_2 as the probe molecule and the TPD profiles of MgO, Ba/MgO, Na/MgO and K/MgO are shown in Fig. 1. Only the MgO shows strong basic sites and the reaction data suggests that these sites are required for the condensation reaction.

Attempts were then made to modify the MgO basicity and/or surface area by addition of acids to the MgO (Table 3). The addition of acids could lead to reaction with MgO ($Mg(OH)_2$) to erode the surface, produce salts, and increase the surface area of the underlying MgO.

The impregnation of MgO with HCl solution yielded mostly butyraldehyde (30%) as the major product, while very small amounts of butanol were observed (0.08%). The phosphoric and acetic acid-loaded MgO catalysts showed slightly improved

Table 3
Activities of acid doped magnesium oxide catalysts for the solid-base condensation of ethanol to butanol[a]

Catalyst	Conversion (mol%)	Yield (mol%)					
		Acetal	Butanal	Crotonal	2-Butanol	Butanol	Crotanol
10% HCl/MgO	22.21	1.93	30.07	0.22	0.08	0.08	0.37
10% H_2PO_4/MgO	42.49	12.02	1.13	0.34	0.33	8.18	0.37
10% Acetic/MgO	21.95	14.28	0.65	0.63	0.36	6.71	0.59
1% H_2SO_4/MgO	49.17	1.65	0.78	1.06	0.17	19.26	1.09
5% H_2SO_4/MgO	38.52	4.42	2.04	1.36	1.27	15.01	0.21
10% H_2SO_4/MgO	50.35	3.19	1.35	0.71	0.85	8.76	0.23
20% H_2SO_4/MgO	58.81	4.16	1.08	0.48	1.04	6.83	0.08

[a] Acetal: acetaldehyde; butanal: butyraldehyde; crotonal: crotonaldehyde; crotanol: crotylalcohol.

results (butanol yield) when compared to the HCl/MgO catalysts. The addition of 1% H_2SO_4 to MgO did not affect the butanol yield significantly. It was anticipated that the neutralisation of the strong basic sites would have led to a change in butanol yield. However, it appears that two effects of H_2SO_4—removal of strong basic sites *and* an increase in catalyst surface area (e.g. 1% H_2SO_4; surface area = $134\,m^2\,g^{-1}$) result in compensatory effects. However, as the sulphuric acid loading on MgO increased, the butanol yield decreased, presumably due to the dominant effect of the poisoning of the strong basic sites on the MgO surface by the large amounts of acid used.

The above study revealed that little advantage was to be gained (activity and/or selectivity) by modification of the MgO with acidic or basic ions. Only a low loading of H_2SO_4 to MgO gave comparable results to that of pure MgO. Further studies thus focussed on the unmodified MgO catalyst.

3.2. Effect of temperature and contact time on the reaction

The effect of reaction temperature on the activity/selectivity of the ethanol dimerisation reaction is shown in Fig. 2. Very little change in the yield of acetal and other C_4 products, as the temperature of the reaction increased, is to be noted. However, there is a very sharp increase in butanol yield at a temperature of 450 °C. Further studies were then performed on the catalyst at 450 °C.

Fig. 3 shows the variation of ethanol conversion and 1-butanol yield with contact time. The conversion of ethanol to products decreased with the increase in contact time. However, the butanol yield stayed fairly constant with increase of contact time, suggesting that it did not decompose over the catalyst via secondary reactions.

A decrease in ethanol conversion was also noted to occur with time on stream (Fig. 4), suggesting that the changes were associated with catalyst decomposition. Again however, the selectivity to butanol was little affected by the change in conversion. The deactivation is assumed to be due to carbon deposition; the catalyst turned black during the reaction. (The catalyst could be reactivated with oxygen at $T = 500\,°C$.)

3.3. Reaction mechanism

Dimerisation of ethanol in the presence of N_2 or O_2 as a carrier gas gave near identical results in terms of

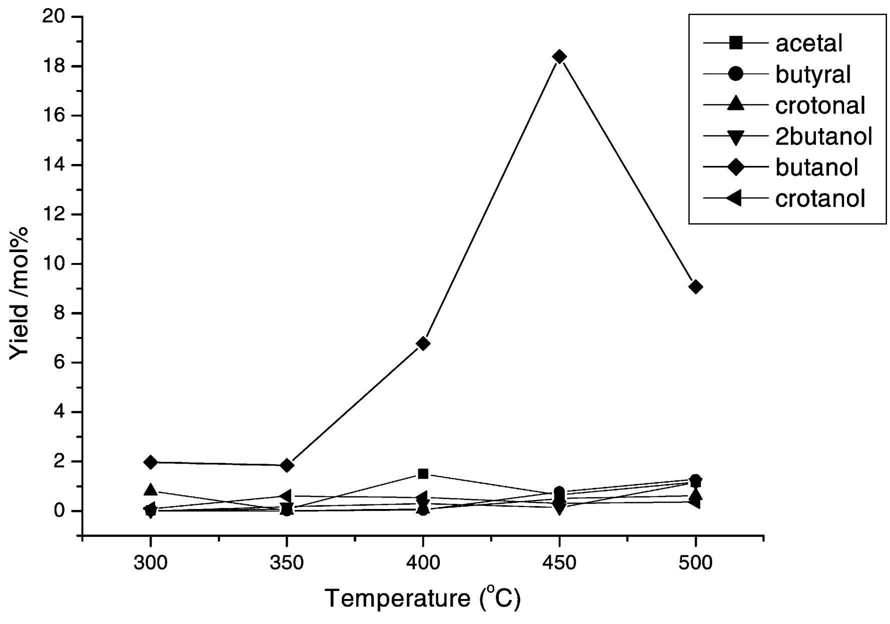


Fig. 2. Influence of temperature on the product distribution over MgO.

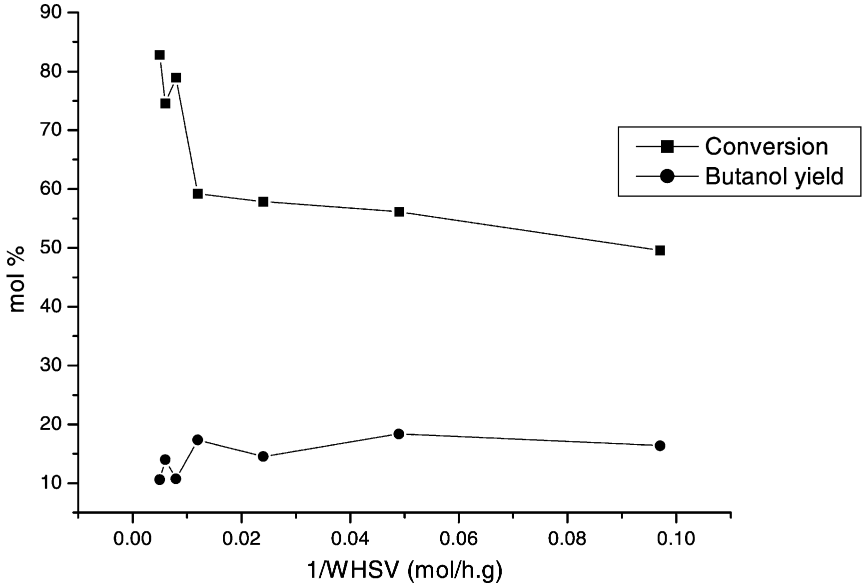


Fig. 3. Influence of contact time on ethanol conversion and 1-butanol yield over MgO.

butanol (and other products) selectivity and activity. Thus, it appears that an oxidizing environment is not necessary for the reaction. An overall reaction scheme has been proposed for the dimerisation of ethanol to propanol by Iglesia and co-workers [13–16] and this is shown in Scheme 1. As can be seen the reaction is proposed to occur via both direct and indirect reactions of ethanol molecules.

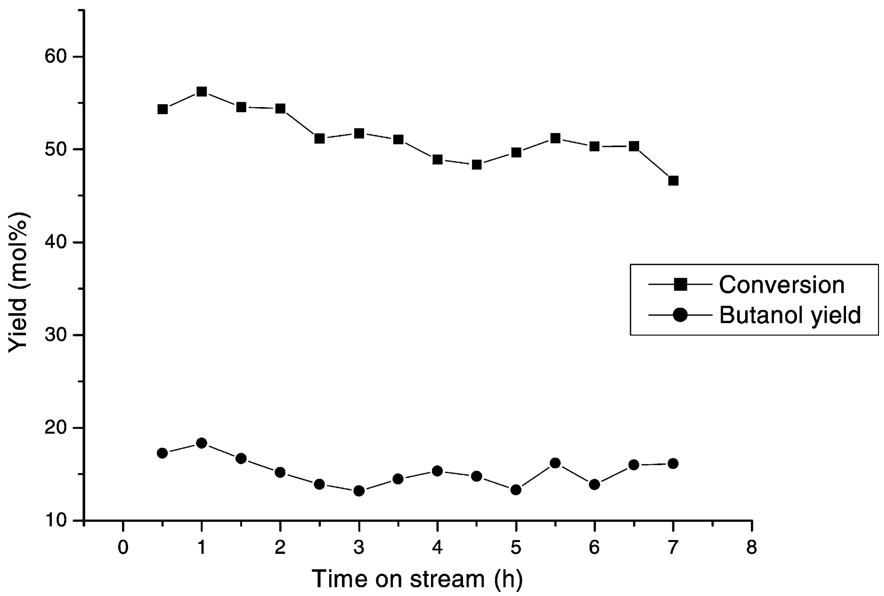
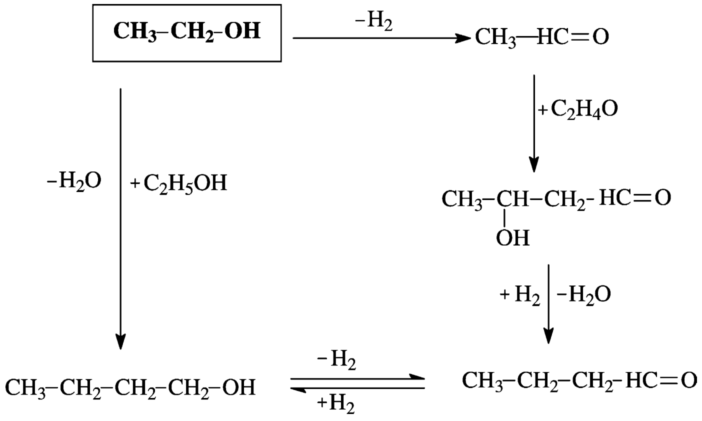


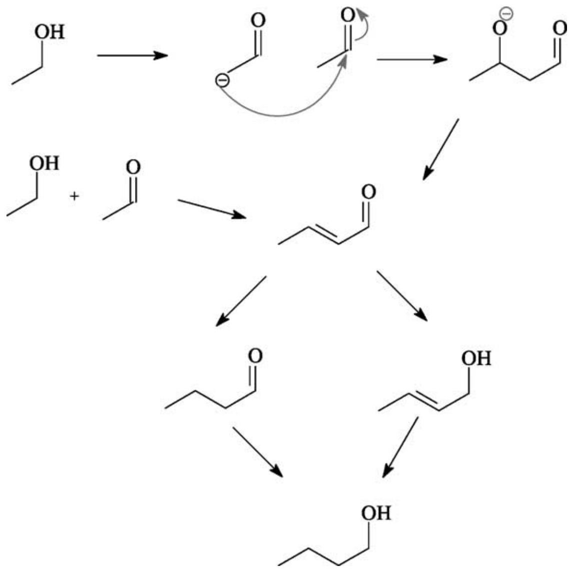
Fig. 4. Influence of time on stream on activity and 1-butanol yield over MgO.



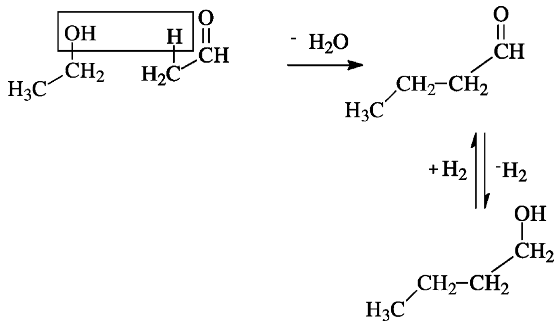
Scheme 1. Mechanism for butanol formation from ethanol.

A number of mechanisms for the production of butanol from ethanol can thus be proposed:

(i) Reaction of ethanol with an alkene via a hydrocarboxylation reaction. This reaction is not considered in this work as little evidence for the presence of ethene was noted (or expected).
(ii) Oxidation of ethanol to acetaldehyde, followed by aldol condensation of acetaldehyde to form crotonaldehyde and finally hydrogenation to 1-butanol (Scheme 2).



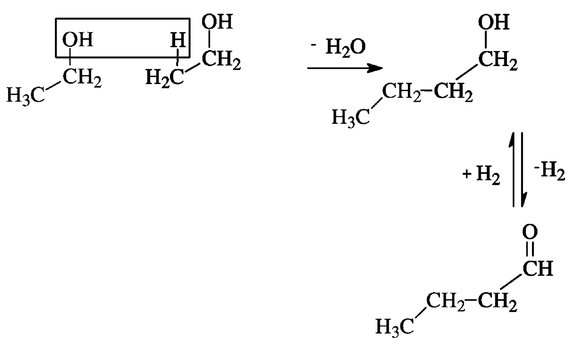
Scheme 2. Aldol condensation mechanism.



Scheme 3. Dimerisation of ethanol and acetaldehyde.

(iii) Dimerisation of ethanol via H abstraction from a β-C atom (Schemes 3 and 4) [19].

To establish the mechanism over MgO, a range of reactions of ethanol with possible reaction intermediates were carried out. Reactions were carried out using



Scheme 4. Dimerisation of two ethanol molecules.

Table 4
The product distribution of the liquid products from the reaction of (i) pure intermediates (100%) and (ii) ethanol/intermediate (1:1) mixtures over MgO[a]

Reagents	Yield (mol%)					
	Acetal	Butanal	Crotonal	2-Butanol	Butanol	Crotanol
Pure reagents						
Acetaldehyde	16.3[b]	4.7	19.1	0.7	5.8	0.4
Butanal	0.1	25.0[b]	0.1	–	15.4	0.3
Crotonaldehyde	0.4	9.4	11.0[b]	0.1	1.6	0.4
Butanol	0.9	13.0	0.2	–	24.5[b]	0.1
Crotylalcohol	1.0	0.9	4.7	–	2.9	20.1[b]
Ethanol	0.3	2.1	0.4	0.5	18.4	0.7
1:1 mixtures[c]						
Acetaldehyde	0.7	4.1	6.4	–	10.1	3.6
Butanal	1.1	10.3	1.5	1.4	40.5	3.6
Crotonaldehyde	0.0	1.3	7.3	–	4.7	0.6
Crotylalcohol	3.4	2.8	–	4.6	7.2	11.9
Butanol	1.9	11.0	1.8	0.7	32.2	3.0

[a] Acetal: acetaldehyde; butanal: butyraldehyde; crotonal: crotonaldehyde; crotanol: crotylalcohol.
[b] Unreacted reagent.
[c] 1:1 mixture of reagent and ethanol.

pure reagents or with the reagents mixed with ethanol (1:1, v/v) (Table 4).

(i) Reaction with pure ethanol provides the base line data as to products formed in the reaction. It is to be noted that very little crotonaldehyde is detected in the reaction.
(ii) Passage of pure butanol or butanal over MgO revealed that the two reagents were readily interconverted. Only minor amounts of other C_4 products were formed. Very little crotonaldehyde was formed from pure butanol, suggesting no possible reaction reversibility via this route. This data is consistent with the findings of Iglesia and co-workers [13–16].
(iii) Passage of acetaldehyde over the MgO did produce some butanol, but also large quantities of crotonaldehyde (19.1%). This suggests that acetaldehyde is not the major intermediate in the mechanism. (This is consistent with the lack of influence of O_2 carrier gas on the reaction.)
(iv) A mixture of ethanol and acetaldehyde gives the expected decrease in crotonaldehyde yield and increase in butanol yield.
(v) Interconversion between crotonalcohol and crotonaldehyde is not significant and crotonaldehyde does not appear to be an intermediate to butanol formation.
(vi) Pure crotylalcohol did not produce 2-butanol.

While some of the results are consistent with an expected aldol condensation reaction, the data suggest a limited role for acetaldehyde as an intermediate in the reaction. Further, while the pure reagents give different results from the ethanol/reagent mixtures, the results are predictable and suggest that reactions of key intermediates required from an aldol reaction play little part in the reaction mechanism.

The data are thus not totally consistent with the mechanism shown in Scheme 2. Acetaldehyde, a by-product, does not contribute to the significant growth of the carbon chain. A key issue relates to the role of hydrogen production/use in the reaction cycle.

A mechanism, shown in Schemes 3 and 4, previously proposed by Yang and Meng [23] for the reaction of ethanol over alkali cation loaded zeolites, is consistent with the data shown in Table 4. In this mechanism, ethanol undergoes proton abstraction from the β-carbon atom and the nucleophilic centre generated attacks another ethanol molecule, resulting in OH displacement (i.e. dehydration), C–C coupling and hence formation of 1-butanol and water. This mechanism is also alluded to in the mechanism proposed by Iglesia and co-workers [13–16] for reactions over basic catalysts.

The TPD data are consistent with this proposal—only strong basic sites would achieve this abstraction from an OH-activated CH_3 group.

Activation of a β-H atom is thus the key step in the mechanism. At a molecular level, it is suggested that ethanol reacts with surface MgO to generate an ethoxide ion and it is this ion that has an activated β-H atom. Further evidence will be required to support this proposal.

4. Conclusions

Our results suggest that acetaldehyde is not a major intermediate in the synthesis of butanol from ethanol. It does participate in a competing coupling reaction (aldol condensation reaction) resulting in the formation of crotonaldehyde. Further, crotonaldehyde is also not an intermediate in the formation of butanol from ethanol.

An alternative coupling reaction proposed by Yang and Meng [23] is possible (Scheme 4), namely—the self-coupling of ethanol. The ethanol/acetaldehyde coupling reaction (Scheme 3) is not consistent with the data and thus the self-coupling of ethanol is the mechanism occurring in this reaction.

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